

If a displayed category has a universal structure, say displayed terminal objects, then if we reindex the displayed category along a functor that preserves terminal objects we should be able to prove that the reindexed category has displayed terminal objects as well.

Intuitively this makes sense, but there is a slight catch practically. That is, we cannot just show that the original category has displayed terminal objects for a fixed terminal object in the base. Instead, we must parameterize by a choice of terminal object in the base and show that we have a displayed terminal object over any terminal structure in the base.

In particular, we have a displayed terminal object in the image of the functor (and thus reindexing witnesses this displayed structure.)